Zoom In, Scale Out: Metric Depth Models Do Not Respect a Digital Zoom
Supplementary Material

Anonymous CVPR submission

Paper ID *****

Section and table numbers without a letter refer to the main paper. Every number here is generated from the same result files as the main paper (`scripts/make_paper_numbers.py`).

A. Qualitative example on NYUv2

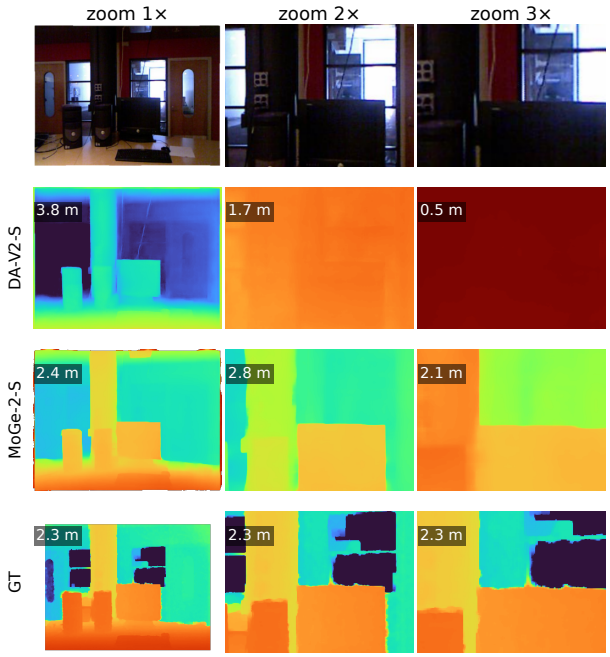


Figure S1. The layout of Fig. 1 on an NYUv2 test image: depth at 1×, 2× and 3× digital zoom on one colour scale in metres. The number in each panel is the median depth over the pixels inside the 3× crop, the same physical points in every column.

B. The depth exponent per dataset

Table S1 and Fig. S2 split the pooled α of Tab. 1 by dataset. On every set each pixels-only model has a larger α than each camera-predicting model (ZoeDepth and MoGe-2-S tie on DIODE-indoor), and Metric3D v2 with the true camera stays near zero. The pixels-only models shrink most on KITTI and NYUv2, where DA-V2-S even exceeds pure apparent-size behaviour ($\alpha > 1$).

Table S1. Depth exponent α per dataset (median over images).

Model	KITTI	NYUv2	DIODE-in	DIODE-out
DA-V2-S	1.28	1.07	0.76	0.71
DA-V2-B	1.19	0.83	0.75	0.66
ZoeDepth	0.82	0.60	0.43	0.52
Depth Pro	0.35	0.46	0.19	-0.01
MoGe-2-S	0.44	0.21	0.43	0.33
UniDepthV2-S	0.60	0.13	0.32	0.25
Metric3D v2-S	-0.03	-0.11	-0.09	-0.18
Metric3D v2-S (stale K)	0.97	0.89	0.91	0.82

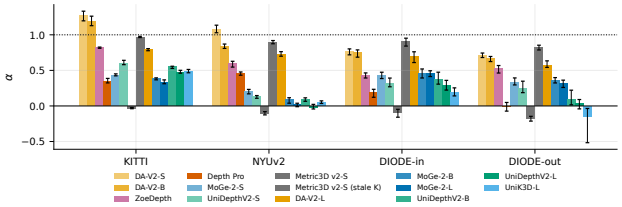


Figure S2. Median α per model and dataset.

C. Off-centre crops and the principal point

Zooms about the four quadrant centres (50 images per set from KITTI and NYUv2) give α of the same order as centred zooms (Fig. S3). For such crops the zoom law moves the principal point by $(c_x - x_0 + \frac{1}{2})\,W/w - \frac{1}{2}$, half the image width at 2×. Figure S4 compares the principal point predicted by MoGe-2 and UniDepthV2 with the law: both errors lie on the curve of a predictor that always answers the image centre (0.496 and 0.494 of the width at 2×, against 0.496).

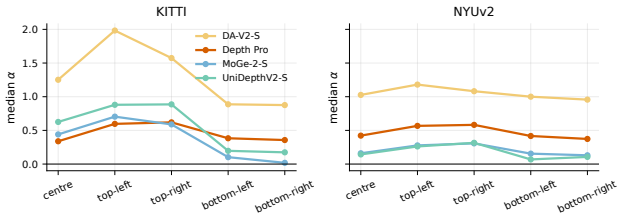


Figure S3. Median α by crop position.

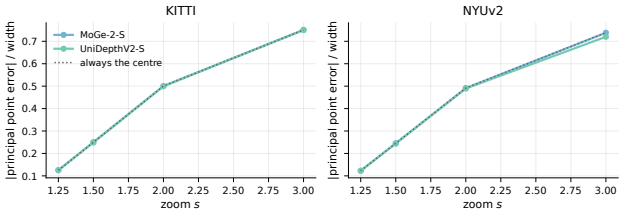


Figure S4. Principal-point error (fraction of the image width) under quadrant zooms. Dotted: the error of always answering the image centre.

D. ZoomFix per dataset

Table S2 gives the cross-fitted ZoomFix of Sec. 5 per dataset. The label-free exponent tracks the dataset: it is above 1 for DA-V2-S on KITTI and NYUv2, where the pooled α hides a stronger shrink. The MoGe-2-S entry for DIODE-outdoor is dominated by a few zoomed images with very large errors (the zoomed AbsRel is 3.037 against 0.379 unzoomed), which a global factor cannot repair.

E. Fine-tuning details

Variants. All variants start from the same DA-V2-S checkpoint and differ only in the loss: EFT (anchor + zoom term), EFT+B (+ blur term), EFT+B with $\lambda = 3$, blur-only (anchor + blur term), and the Zhong *et al.*-style target (anchor + zoom term, with the student's own stop-gradient prediction on the unzoomed image as target). Each is trained once per domain (KITTI raw outdoor, NYUv2 train indoor) with the settings of Sec. 6.

Training curves. Figure S5 shows the zoom and anchor terms over training (running mean over 100 steps). The anchor term is exactly zero at the first step, since the student is a copy of the teacher; it jumps as soon as the zoom term moves the weights and then falls through training (EFT outdoor: mean 0.10 over the first 100 steps, 0.04 over the last 100), as the model learns to satisfy both terms.

F. Models and compute

Checkpoints. Hugging Face ids unless noted.

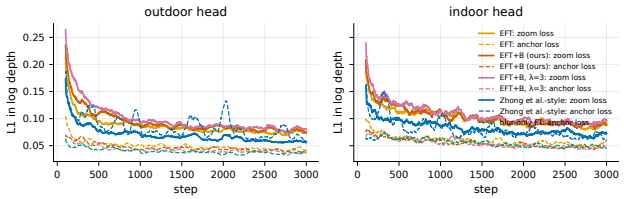


Figure S5. Zoom loss (solid) and anchor loss (dashed) of the fine-tuning variants, outdoor and indoor heads. Blur-only has no zoom term.

- DA-V2: `depth-anything/Depth-Anything-V2-Metric-{Indoor,Outdoor}-{Small,Base,Large}-hf`
- ZoeDepth: `Intel/zoedepth-nyu-kitti`
- Depth Pro: `apple/DepthPro-hf`
- MoGe-2: `Ruicheng/moge-2-{vits,vitb,vitl}-normal`
- UniDepthV2: `lpiccinelli/unidepth-v2-{vits14,vitb14,vitl14}`
- UniK3D: `lpiccinelli/unik3d-vitl`
- DA3: `depth-anything/DA3NESTED-GIANT- LARGE-1.1`, `DA3METRIC-LARGE`, `DA3-LARGE-1.1`
- VGGT: `facebook/VGGT-1B`
- Metric3D v2: `metric3d_vit_{small,large}` (official torch hub entry)

Compute. All audits and fine-tuning runs ran as Kaggle notebooks with one GPU each; the analysis runs on a laptop CPU. The code release contains the kernel builders, the run logs and every result CSV, and `scripts/verify_results.py` checks that all models saw the same images and all zoom rows are complete.

Table S2. ZoomFix per dataset at $3\times$, cross-fitted: the exponent $\hat{e}$ is measured without labels on half of the images and the correction scored on the other half. AbsRel of the zoomed prediction as is, after multiplying by $s^{\hat{e}}$, and of the unzoomed prediction on the same pixels (the target).

	KITTI				NYUv2				DIODE-in				DIODE-out			
Model	$\hat{e}$	as is	fixed	target	$\hat{e}$	as is	fixed	target	$\hat{e}$	as is	fixed	target	$\hat{e}$	as is	fixed	target
DA-V2-S	1.25	0.758	0.384	0.149	1.12	0.675	0.495	0.205	0.75	0.598	0.365	0.263	0.69	0.558	0.661	0.605
Depth Pro	0.31	0.258	0.277	0.160	0.44	0.449	0.223	0.121	0.19	0.399	0.421	0.201	0.12	0.636	0.638	0.665
MoGe-2-S	0.44	0.536	0.262	0.255	0.12	0.272	0.227	0.129	0.44	0.493	0.364	0.257	0.38	3.037	4.396	0.379